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Is Schumpeterian ‘creative destruction’ a plausible source of endogenous real business cycle shocks?

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Abstract

This paper examines the link between growth and business cycles by incorporating Schumpeterian growth into an otherwise standard real business cycle (RBC) model. We ask whether Schumpeter’s creative destruction process can generate realistic business cycles. We solve the model, generate artificial time series, and compare statistical properties of these series with their historical counterparts.

In our approach, the trend component is included in our artificial series just as it is in the data. Hence, we are not tied to any particular filtering method for our comparisons.

We find that Schumpeterian fluctuations alone can generate time series that match key moments from U.S. data. We also find, however, that a model with both Schumpeterian and standard RBC shocks performs better.

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1. Introduction

Dynamic general equilibrium models have proved to be valuable tools for examining both economic growth and fluctuations. One class of these models – the Schumpeterian or ‘quality ladders’ models – focuses on explaining observed smooth growth trends. Segerstrom et al. (1990), Grossman and Helpman (1991a,b), and Aghion and Howitt (1992) first introduced this literature in their seminal papers. In addition, there have been numerous extensions of the basic quality ladders model, focusing on innovation versus imitation, North–South trade patterns, and other related topics.¹

Schumpeterian models have the advantages that they are rigorously based in microeconomic theory and also have intuitive appeal. Growth is driven endogenously by attempts to innovate and climb up the quality ladder to capture a stream of monopoly profits. Attention is normally focused on the steady state, where growth is smooth over time due to a large number of independent, but identical innovators each targeting a unique good. Growth for any given good proceeds in a lumpy fashion with discrete jumps in quality occurring randomly over time, but the law of large numbers leads to smooth aggregate growth.

A second class of models focuses on explaining the behavior of economic aggregates over the course of the business cycle. Usually referred to as real business cycle (RBC) models, early work began with seminal papers by Kydland and Prescott (1982) and Long and Plosser (1983). Again, numerous papers have extended this literature to the examination of aggregate labor behavior, monopolistic competition, monetary aggregates, and various other topics.²

The RBC methodology generally involves building a general equilibrium model, with changes in productivity (and, more recently, other economic fundamentals) driving aggregate behavior. These models are capable of generating artificial data that mimic observed business cycles. Unlike the Schumpeterian literature, attention is focused on the off-steady-state high-frequency behavior of aggregate economic variables.

In practice, RBC models are typically transformed into a stationary variant and solved numerically to yield stationary laws of motion for endogenous variables as functions of endogenous states and exogenous driving processes. A stationary model economy is then simulated and evaluated by comparing properties of data drawn from the model with data drawn from actual economies. Since the focus of RBC models is on high-frequency fluctuations and not on economic growth, the growth component is usually ignored. However, in order to compare the artificial economy with real world data it is necessary to remove the growth component from the real world data; i.e. to detrend it. While many business cycle stylized facts are invariant to the filter used, there are some important facts that are not.³

¹For an extensive overview of this literature, see Grossman and Helpman (1991a, b) and Aghion and Howitt (1998).

²For an excellent overview of the RBC literature, see Cooley (1995).

³See Canova (1998).

Generally in RBC models, the source of shocks is an exogenously imposed sequence of large and volatile productivity shocks. A common specification of such shocks is a simple AR(1), and there is little or no economic theory involved in the specification of the driving process. Moreover, there is increasing skepticism that technology shocks, measured by Solow (1957) residuals, are a major source of business cycle fluctuations. As King and Rebelo (2000) point out “A key difficulty is that typical estimates of Solow residuals imply a probability of technical regress on the order of 40%, which seems implausible to most economists.”

This paper integrates the two branches of literature identified above: the RBC literature, which focuses on detrended, high frequency fluctuations; and the Schumpeterian literature, which focuses on low-frequency growth trends.

In our model, as in all Schumpeterian or quality-ladders models, growth is the result of research and development activities which use scarce factors (labor in our case) to attempt to discover a new method of producing one of many intermediate goods. In much of the literature the new technology yields a higher quality good. However, we model the new technology as a more efficient method of producing an existing good which never varies in quality.

Mathematically, the two approaches are identical. Firms are willing to hire R&D resources now in exchange for a stream of future monopoly profits that result from being able to undercut the current producer once the new technology is in hand. The continual pursuit of these profits across industries and over time drives growth in our model. Since there are private incentives to engage in R&D, we ignore any role the government might play in providing R&D directly or providing incentives for others to pursue it. These have been addressed in other work and are not the focus of this paper.

The objective is to construct a dynamic general equilibrium model with an endogenous driving process for business cycles derived from microeconomic primitives, and with high- and low-frequency movements in economic aggregates that mimic those observed in the U.S. economy. The endogenous growth component of the model is included in simulations, which create artificial data. Properties of the artificial data are then compared to like properties of their real data counterparts to evaluate the model's performance.⁴

An important feature of the model is that growth and cyclical properties of data stem from a common source – endogenous innovations to technology. Technical regress is not necessary for business cycle fluctuations in the model. Rather, the endogenous movements of resources between goods production and technological advancement gives rise to cyclical fluctuations, as well as lower frequency movements in key macroeconomic variables. In principle, the model does not require large and variable technology shocks that imply high likelihoods of technical regress to explain business cycles. Rather, endogenous improvements in technology and the diffusion of the improvements into production of final goods can help explain both growth and cycles.

⁴Note that since growth is endogenous in the model, the choice of which particular detrending method to use is less critical than in a prototypical RBC model.

As Aghion and Howitt (1992) point out, a model with a finite number of R&D sectors may have equilibria beside the steady or stationary state. These sunspot equilibria may well generate cyclical behavior, but our model focuses attention on cycles driven by the stochastic arrival of new technologies and not on behavior driven by self-fulfilling expectations.

1.1. Methodology

Our methodology is as follows. A small number of industries are assumed⁵; giving rise to aggregate growth in technology that is ‘lumpy’ and can therefore serve as a driving process for business cycles.

The model is evaluated by the following: linearizing agents’ Euler equations, with market clearing conditions imposed; numerically solving the model for endogenous variables as functions of endogenous states and exogenous shocks; simulating the model to generate sequences of macroeconomic aggregates; comparing properties of data generated by the model with properties of data drawn from the U.S. economy at both high- and low-frequencies.

There are two sources of shocks in the model. One (denoted A) is a sequence of random draws to determine success or failure of potential innovators who invest resources in basic research and development (R&D) to influence their success probabilities. In aggregate the size of this shock goes to zero as the number of independent industries assumed approaches infinity. A second source of shocks (denoted z) is innovations to labor productivity governed by a process typically used in RBC models. This set-up has the advantage of nesting both the pure RBC model and the Schumpeterian model as special cases. To examine the dynamics of the RBC model we can set the number of intermediate sectors to a very large number (like 1,000,000) and virtually all fluctuations will come from the exogenous shocks. Similarly, we can set the variance of the exogenous shocks to zero and the number of industries to a small number to examine the dynamics of the pure Schumpeterian case.

1.2. A comparison with other dynamic Schumpeterian models

Our approach is related to papers by Fan (1995), Ozlu (1996), Andolfatto and MacDonald (1998), Collard (1999), Freeman et al. (1999), Wälde (2002, 2005) and Maliar and Maliar (2004) which also focus on dynamic implications of endogenous growth models. There is also an expanding literature on ‘Schumpeterian waves’, which looks at the behavior of the economy in response to large but infrequent movements in basic technology. Early work by Cheng and Dinopoulos (1992, 1996) looked at these longer-run fluctuations. Aghion and Howitt (1998) devote a portion

⁵Alternatively, we could imagine a large number of industries partitioned into a small number of sectors. Suppose innovations spill over perfectly within the sector, i.e. any innovation is enjoyed by all firms in all industries within the sector regardless of which firm does the actual innovation. This setup will be equivalent to simply having a small number of industries.

of their book to this phenomenon as well.⁶ Wälde (2002) examines the implications of Schumpeterian growth in a continuous time framework, but does not calibrate or compare model generated data with real world data.

Collard (1999) and Ozlu (1996) each consider extensions of a standard RBC model to include endogenous growth through human capital accumulation effects of learning-by-doing. Ozlu considers labor market implications of allowing learning-by-doing effects on human capital, while Collard considers implications for the autocorrelation of output growth and impulse response functions in the trend-reverting component in output. These authors find improvements in quantitative implications of RBC models augmented to include learning-by-doing effects over standard RBC models that lack human capital features. Our analysis is similar to those performed by Collard and Ozlu in two respects. First, we also consider business cycle implications of an RBC model augmented to include an endogenous growth mechanism. Second, as in Collard's analysis, we consider implications of our model for fluctuations in key variables at various frequencies including, but not limited to, business cycle frequencies. Our model differs from theirs, however, in an important way. While Collard and Ozlu essentially provide a model of an endogenous mechanism for the propagation of exogenous technology shocks, we model *both* propagation of shocks and the shocks themselves. That is, we present a model that accounts for how shocks to technology arise, as well as how they may be propagated and diffused through time.

Closer to the spirit of our analysis is the work by Andolfatto and MacDonald (1998), and by Freeman et al. (1999). The latter set of authors construct a model of large and costly technological changes which give rise to deterministic cycles and long run growth. Their economy requires a sufficiently large amount of capital, diverted from consumption and physical investment, for birth of a technological innovation. Then, when an innovation occurs, capital is more highly valued in physical investment than in R&D investment. Consequently, resources flow away from R&D and toward final goods production. As the marginal product of capital using existing technology fades through time, resources subsequently flow back toward consumption and the production of R&D innovations. These flows of capital give rise to endogenous movements in consumption and investment patterns within each *fixed-length* innovation cycle. Our analysis similarly accounts for endogenous movements in key macroeconomic variables within innovation cycles, but also explicitly models innovation cycles of *random* durations.⁷ In addition, when exploring quantitative predictions of our model relative to quantitative properties of actual data, we explicitly consider movements in key macroeconomic variables at well-defined frequencies. In contrast, Freeman et al. are agnostic about the frequency one should look at in actual data for the innovation-driven cyclical patterns of movements that their model predicts.

⁶Chapter 8, especially Section 8.4.

⁷Another difference between our model and the one constructed by Freeman et al. is that our model considers labor, rather than capital, as the input to the R&D process. We include only labor to simplify the analysis but are free to enter capital into the R&D process as well.

The analysis of Andolfatto and MacDonald is close to that in this paper in that they consider fluctuations and growth in key macroeconomic aggregates arising from the discovery and diffusion of technological innovations. As in our model, Andolfatto and MacDonald have growth arising from technological discovery and use, and fluctuations arising from diffusion of applied, or frontier, research. Some type of diffusion mechanism is required in each model to smooth out what would otherwise be unrealistic spikes in economic aggregates from infrequent, possibly large, technological innovations springing up from applied research. They use a diffusion mechanism involving imitation and learning how to use new technology, which diverts resources from production. Agents choose the amount of resources to devote to various imitation and learning possibilities available to them, the outcomes of which are random. With such a mechanism, the authors compare Hodrick–Prescott (HP) filtered series (in levels, not deviations from filtered series) from actual data with like series drawn from their model.

The model in this paper has no such mechanism for innovation and diffusion, though previous work has focused on a similar diffusion process. Our analysis differs from Andolfatto and MacDonald's in the frequencies of movements in macroeconomic variables considered. Our model is an attempt at simultaneously accounting for growth and business cycle fluctuations. Consequently, we consider movements in macroeconomic variables at various frequencies, including those that arise at business cycle frequencies. We do not restrict attention to data movements at frequencies at or below those at which major innovations diffuse, they do.

The models closest to ours are those of Fan (1995), Wälde (2002, 2005), and Maliar and Maliar (2004). Fan (1995) considers representative agents who choose the optimal length of time to wait for R&D projects to come to fruition. Inventions arrive in a random manner, but are not immediately implementable. Imperfectly substitutable time and labor effort are required to transform these inventions into usable technology. High quality inventions impose higher opportunity costs of waiting to implement and thus lead to shorter R&D cycles. As we do, Fan also allows for short-run productivity fluctuations in the RBC tradition to drive short-run fluctuations in output. Unlike our paper, however, Fan is interested in endogenous technical progress as a way of explaining long and medium-term movements and views it as having little to do with movements at business cycle frequencies.

Wälde (2002) build a model with many of the same features as ours. The major differences are that he works in continuous time, while we use a discrete-time framework. In his model, some fraction of capital depreciates due to obsolescence when an innovation occurs, which allows the model to generate substantial downturns in output. Cycles involve adjustment periods after an innovation, during which the capital stock is built up and R&D activities are shut down. Wälde provides analysis of the steady state and the adjustment path to the steady state, but does not simulate the model numerically and compare it to historical time-series.

Maliar and Maliar (2004) build a similar model with a Poisson arrival process for new technologies. Technology arrival probabilities are functions of the aggregate levels of human capital (which is used solely for innovation) and physical capital.

Hence, technical progress is the result of general purpose advances which impact the whole economy. In contrast, we focus on technical progress that is driven by deliberate attempts to capture monopolistic profits and which directly impact only a fraction of the economy. Intriguingly, they find that some parameterizations of their model can match the persistence and asymmetries of output growth for the U.S., without relying on exogenous shocks that are highly autocorrelated. We consider a narrower range of parameter values and have difficulty matching output asymmetry.

2. A dynamic general equilibrium Schumpeterian model

In this section, insights of the Schumpeterian or ‘quality ladders’ growth models are incorporated into a discrete-time stochastic general equilibrium model of the RBC tradition. Since it is well known that the equilibrium from Schumpeterian growth models is socially suboptimal, we proceed to examine the competitive equilibrium. First, we examine the behavior of households, then that of production firms, and finally the behavior of research firms. Imposing aggregate resource constraints and market clearing conditions closes the model.⁸

The model contains households, production firms, and research firms. Each infinitely lived household is endowed with one unit of labor each period supplied to firms at wage w . Households also accumulate physical capital, K , over time, which they rent to firms each period at rental rate r . In addition to physical capital, households buy and sell equity shares in two types of existing firms—production and research firms—in I different intermediate sectors. These shares influence the household’s budget by generating dividends and capital gains or losses. There is also a final goods sector, which we assume is perfectly competitive and generates no profits. For simplicity, we abstract from buying and selling of these firms’ equities.⁹

In each period there is a single production firm in each intermediate industry i with an exclusive right to a particular level of production technology, A_i , that is some factor $\theta > 1$ better than the closest competitor. This production firm enjoys monopoly power and earns monopoly rents during the current period, and possibly many future periods, until a firm with even better technology replaces it. The production firms hire labor and rent capital to produce intermediate goods, and pay out profits as dividends to shareholders each period. Final goods are produced by combining intermediate goods.

There also exists a single new research firm for each intermediate industry, which incorporates with the intent of displacing the current production firm in its role of monopolist. The research firm issues equity shares and uses the proceeds to hire units of labor to attempt an innovation. If successful, the research firm discovers a technology that is a factor θ better than the current production firm and begins production as the monopolist next period. If unsuccessful, the firm ceases to exist and its equity shares become worthless. We account for successes and failures across

⁸A listing of notation and variables used in the model is contained in Table 1.

⁹The assumption is innocuous because these equities would have zero prices.

Table 1
Definition of variables

Endogenous variables that change over time	
A_i	level of applied knowledge in intermediate industry i
A	aggregate level of applied knowledge
z	aggregate RBC-style productivity shock
K	capital stock owned by household
K_i	capital employed in industry
P_i	shares of production firm i owned by household (1 in equilibrium)
R_i	shares of research firm i owned by household (1 in equilibrium)
S_i	state of research success for industry i ; 1 is success, 0 is failure
w	real wage
r	real interest rate
L_i	labor hired by research firm i (same for all i in equilibrium)
L	aggregate labor hired by all R&D firms
N_i	labor hired by production firm i (same for all i in equilibrium)
N	aggregate labor hired by all production firms
Y_i	output of intermediate good i
Y	output of final goods
p_i	price of intermediate good i
π_i	profits earned by current production firm i (same for all i in equilibrium)
q_i^P	price of one share in the current production firm i (same for all i in equilibrium)
q_i^R	price of one share in the current research firm i (same for all i in equilibrium)
ρ_i	probability that $S_i' = 1$ (same for all i in equilibrium)
I	number of industries in the economy
J	number of industries that successfully innovate; $J \leq I$
η	random innovations to z
Parameters	
α	capital share in output from a Cobb–Douglas production function; $0 < \alpha < 1$.
β	time discount factor; $\beta < 1$.
γ	CES parameter from momentary utility function; $\gamma > 0$.
δ	rate of depreciation; $\delta > 0$.
θ	growth factor for A_i when $S_i = 1$; $\theta > 1$.
κ	sensitivity of ρ to R&D inputs; $\kappa > 0$.
ψ	autocorrelation parameter for z .
σ^2	variance of innovations in z .

industries with an I -dimensional vector S' , with element $S_i' = 1$ if a research firm succeeds in the current period and $S_i' = 0$ if not. If $S_i' = 1$, then today's research firm in industry i becomes the production firm tomorrow with technology A_i' that is a factor θ better than its predecessor.¹⁰

There is also an aggregate random shock to productivity in the model, unrelated to any of the A_i 's which we denoted z . This exogenous shock is to the productivity

¹⁰We adopt the notation convention that variables without a prime denote current period values and variables with a prime denote next-period values. Additionally, bold letters are used to denote I -dimensional vectors of variables.

of labor in the intermediate goods producing firms' production functions. We include such shocks to allow for shocks to productivity unrelated to actual movements in technology, such as oil price shocks, changes in marginal tax rates, changes in government regulation of production processes, or similar non-technology shocks. This shock to productivity, assumed to be common to all I intermediate good producers, is of the form used in standard RBC models. One of our interests will be the extent to which, by allowing for random innovations in R&D, the model does not require productivity shocks that are as persistent and volatile as those typically employed in RBC models to explain business cycle fluctuations.

The timing of information, shocks, and activities in the economy is as follows. Agents begin a period with capital stock K knowing research and shock realizations $\mathbf{A}$,¹¹ and z . At the beginning of the period, factor and equity markets open and clear. Firms rent capital from households, and real rental rate r is determined. Firms also hire labor from households, and real wage w is determined. Research firms issue shares, and prices of those shares, $\mathbf{q}^R$ (also a vector), are determined, and production firms issue shares, and their prices, $\mathbf{q}^P$, are determined. Following input and funding acquisitions, production of goods and research occurs. Once these activities have been completed factor payments w and rK are made, production-firm profits π are distributed to shareholders, next period's capital stock K' is chosen, and consumption occurs. At the end of the period the research results and random shocks $\mathbf{A}'$, and z' are revealed. We now turn to decisions made by the household, production firms, and research firms.

2.1. The household's problem

A representative household enters a period with capital K carried over from the previous period and a normalized unit labor endowment.¹² The household also owns stocks of equity shares in last period's production and research firms, denoted by share vectors $\mathbf{P}$ and $\mathbf{R}$. The household knows the following: the current levels of technology, $\mathbf{A}$, to be employed by this period's production firms, the current random productivity shock, z , and whether last period's research firms succeeded or failed, the vector $\mathbf{S}$. Taking prices r , w , $\mathbf{q}^P$, $\mathbf{q}^R$, and the probabilities of success by the current research firms, $\mathbf{p}$, as given, the household chooses new stocks of equities $\mathbf{P}'$ and $\mathbf{R}'$ to carry over to next period.

After production is completed and next period's values for technology, $\mathbf{A}'$, are revealed, the household chooses a level of capital, K' , to carry into next period. Consumption then occurs and tomorrow's value for the exogenous shock, z' , is revealed.¹³

¹¹This is a vector containing all the A_i shocks

¹²The household supplies its endowment inelastically and therefore receives wages $w * 1$. The labor endowment is divided between research and production firms' activities.

¹³This odd timing is due to the need to know shocks to A when computing decision rules for the stationary model where $\hat{K}' = K'/A'$.

The value function for the household is

$$V(K, \mathbf{P}, \mathbf{R}; \mathbf{\Omega}) = \text{Max}_{P', R'} E \left\{ \text{Max}_{K'} [u(C) + \beta V(K', \mathbf{P}', \mathbf{R}'; \mathbf{\Omega}')] \right\},$$

where $C = w + (1 - \delta + r)K - K' + \sum_{i=1}^I \{[\pi_i + q_i^P][(1 - S_i)P_i + S_i R_i] - q_i^P P'_i - q_i^R R'_i\}$, $u(C)$ is the momentary utility function, β is the discount factor, $\mathbf{\Omega} = \{w, r, \mathbf{q}^R, \mathbf{q}^P, \mathbf{p}, \mathbf{S}, \mathbf{A}, z\}$ is an information set, and E is the expectation operator given information available at the beginning of the period. $\mathbf{p}$ in the information set represents the vector of industry R&D success probabilities. $\mathbf{S}$ represents the vector of industry R&D success indicator variables, taking values of 1 if a success occurs, and 0 if failure occurs.

Note that the consumer's problem does not allow for a labor–leisure choice. Thus, in contrast to standard RBC models, ours has nothing to say about the cyclical comovements of employment with the other relevant macro variables.

For industry i , ρ_i is the probability of a research firm successfully innovating in the current period. With probability ρ_i a share in today's industry i research firm will pay off next period. If today's research firm pays off next period, then a share in that industry's current intermediate goods producing firm will not pay off next period because it is replaced by the current period's successful innovator. Correspondingly, with probability $1 - \rho_i$, industry i 's current research firm is unsuccessful and will not pay off next period. In that case, the current intermediate goods producer remains as next period's producer providing payoffs on its shares.

The envelope conditions from the household's problem consist of I conditions each for the shares P_i and R_i , and a condition for capital stock K given, respectively, by

$$V_{P'}^i(K, \mathbf{P}, \mathbf{R}; \mathbf{\Omega}) = E\{u_C(C)(\pi_i + q_i^P)(1 - S_i)\}, \quad i = 1, 2, \dots, I,$$

$$V_{R'}^i(K, \mathbf{P}, \mathbf{R}; \mathbf{\Omega}) = E\{u_C(C)(\pi_i + q_i^P)S_i\}, \quad i = 1, 2, \dots, I,$$

$$V_K(K, \mathbf{P}, \mathbf{R}; \mathbf{\Omega}) = E\{u_C(C)(1 - \delta + r)\}.$$

The Euler equations corresponding to the household's choices consist of I conditions each for next period shares P'_i and R'_i , and a condition for next period's capital stock K' given by

$$E\{u_C(C)\}(-q_i^P) + \beta E\{V_{P'}^i(K', \mathbf{P}', \mathbf{R}'; \mathbf{\Omega}')\} = 0 \quad \text{for } i = 1, 2, \dots, I,$$

$$E\{u_C(C)\}(-q_i^R) + \beta E\{V_{R'}^i(K', \mathbf{P}', \mathbf{R}'; \mathbf{\Omega}')\} = 0 \quad \text{for } i = 1, 2, \dots, I,$$

$$E\{u_C(C)\}(-1) + \beta E\{V_K^i(K', \mathbf{P}', \mathbf{R}'; \mathbf{\Omega}')\} = 0.$$

Combining envelope and Euler equations gives the following $2I + 1$ system of equations:

$$E\{u_C(C)\}q_i^P = \beta(1 - \rho_i)E\{u_C(C')(\pi'_i + q_i^{P'})|A_i = \theta\}, \quad i = 1, 2, \dots, I, \quad (1)$$

$$E\{u_C(C)\}q_i^R = \beta\rho_i E\{u_C(C')(\pi'_i + q_i^{P'})|A_i = 1\}, \quad i = 1, 2, \dots, I, \quad (2)$$

$$u_C(C) = \beta E\{u_C(C')(1 - \delta + r')|\mathbf{A}\}, \quad (3)$$

where expectation operator $E\{x|y\}$ denotes the expectation of x given all the information available at the beginning of the period, *plus* additional information revealed after the beginning of the period that is contained in y .

The laws of motion governing each industry's technology level A_i , research success index S_i , and exogenous technology shock z_i , known to the household, are

$$A'_i, S'_i = \begin{cases} \theta A_i, 1 & \text{with probability } \rho_i, \\ A_i, 0 & \text{with probability } 1 - \rho_i, \end{cases} \quad (4)$$

$$z' = \psi z + \eta'; \quad \text{where } \eta' \text{ is distributed Normal } (0, \sigma^2). \quad (5)$$

The endogenous choices and random shocks governing the R&D success probabilities ρ_i are discussed in detail below when the R&D firms' choice problems are discussed. The z shocks are the productivity shocks discussed earlier.

2.2. Production of final goods

Final goods production is an Armington aggregator of all intermediate goods, and uses no capital or labor¹⁴:

$$Y = I \prod_{i=1}^I Y_i^{1/I}. \quad (6)$$

Firms view the prices of intermediate goods as fixed, and a typical firm maximizes profits:

$$\Pi_F = I \prod_{i=1}^I Y_i^{1/I} - \sum_{i=1}^I p_i Y_i. \quad (7)$$

First-order conditions for the firm's problem yield

$$p_i Y_i = \frac{Y}{I}. \quad (8)$$

This shows that all intermediate goods producers earn the same amount of real revenue. Alternatively stated, regardless of the amount produced, expenditures on each intermediate good are equal.

2.3. Production of intermediate goods

The production firm produces output using a Cobb–Douglas production function with two sources of productivity variation, both of which are assumed to be labor augmenting:

$$Y_i = K_i^\alpha [e^z A_i N_i]^{1-\alpha}. \quad (9)$$

¹⁴We could take the usual approach and use this aggregator as a utility function expressing a preference for variety across the I intermediate goods. Interpreting it as a final good, however, has the advantage of yielding a natural numeraire good for the calculation of real values.

Productivity variations come from productivity shocks, z , along with the endogenous growth shocks A_i .

The monopolistic production firm faces a downward sloping demand curve defined by (8). However, there is a potential competitor that places limits on the price the monopolist will charge. The previous producer of the good has access to a technology that is $1/\theta$ as productive as the current firm's. The current firm will never charge a price that exceeds its marginal cost by more than a factor of $\theta^{1-\alpha}$. To do so would be to surrender production to the previous producer. It can easily be shown that the marginal cost for the current firm is:

$$\left(\frac{1}{e^z A_i}\right)^{1-\alpha} \left[\left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha}\right)^\alpha \right] w^{1-\alpha} r^\alpha.$$

Similarly, the marginal cost for its closest competitor is:

$$\left(\frac{\theta}{e^z A_i}\right)^{1-\alpha} \left[\left(\frac{\alpha}{1-\alpha}\right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha}\right)^\alpha \right] w^{1-\alpha} r^\alpha.$$

Hence, the optimal price for the firm is to charge a multiplicative markup of $\theta^{1-\alpha}$ over marginal cost.

Standard optimality conditions for the firm reveal that it divides revenues between payments to labor, capital, and dividends according to

$$wN_i = \frac{1-\alpha}{\theta^{1-\alpha}} p_i Y_i = \frac{1-\alpha}{\theta^{1-\alpha} I} Y,$$

$$rK_i = \frac{\alpha}{\theta^{1-\alpha}} p_i Y_i = \frac{\alpha}{\theta^{1-\alpha} I} Y,$$

$$\pi_i = \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha}} p_i Y_i = \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha} I} Y.$$

These distribution equations show that capital, labor, and profits are identical for all intermediate firms regardless of the level of technology they use. The aggregate values are, then, $K = IK_i$, $N = IN_i$ and $\pi = I\pi_i$. Substituting these values into (9) and (6), the production of final goods can be written as an aggregate production function of the form

$$Y = I \prod_{i=1}^I [K_i^\alpha (e^z A_i N_i)^{1-\alpha}]^{1/I} = K^\alpha (e^z A N)^{1-\alpha}, \quad (10)$$

where

$$A \equiv \prod_{i=1}^I A_i^{1/I}. \quad (11)$$

This shows how we can collapse the production of intermediate and final goods into a single aggregate production function.

Given the manner in which the A_i evolve in (4), and the aggregation in (11), aggregate technology, A , evolves according to the following Binomial law of motion:

$$A' = A\theta^{J/I}; \quad \text{with } J \text{ distributed as Binomial } (I, \rho) \quad (12)$$

2.4. The research firm's problem

Each period, a single research firm springs into existence in each industry. It sells equity shares, normalized to a quantity of one, to the household at price q_i^R and uses the proceeds to hire labor. Taking prices w and q_i^R as fixed, the firm chooses the amount of labor to hire by solving

$$\text{Max}_{L_i} \Pi_R = \rho_i E \left\{ \frac{V'_i}{1+r} | S' = 1 \right\} \quad \text{s.t. } wL_i \leq q_i^R, \quad (13)$$

where $V'_i = \pi'_i + q_i^{P'}$ is the reward for a successful innovation, and ρ_i is the probability of success.

The reward for success consists of the expected present value of the stream of profits given that a success occurs, which happens with probability ρ_i . Appendix A shows that this reward can be written as

$$V'_i = \frac{\theta^{(1-\alpha)} - 1}{\theta^{(1-\alpha)}} \sum_{s=t+1}^{\infty} \frac{1}{I} Y_s d_{i,s} \quad (14)$$

where $d_{i,s}$ is a discount factor defined in the appendix. The right side of Eq. (14) is simply the discounted sum of all future profits in the industry, with discounting inclusive of time and probabilities of loss of the profit stream in future periods.

We model the research firm as hiring labor inputs that are used to produce research tries. If each unit of R&D draws either a success or failure from the same distribution, the firm's overall probability of success will come from a Binomial distribution. In the limit, with a continuous measure for the tries, the probability of success comes from a Poisson distribution. Multiple successes are assumed to produce identical technology, so the firm's overall probability of success is the probability that at least one unit succeeds. This is

$$\rho_i = 1 - \exp\{-\kappa L_i\}, \quad (15)$$

where κ is the 'ease' of doing research.¹⁵

Eq. (15) reflects the notion that all R&D units operate independently, i.e. that the R&D process is expanded by duplicating existing activity. When producing goods this would imply constant returns to scale. In this context the output is a probability and is bounded above at one. Duplicability here does not imply homogeneity of

¹⁵As the length of the period goes to zero this converges to the linear probability function found in the continuous time case. To see this substitute take the derivative of (15) with respect to L_i and substitute $L_i t$ for L_i to see what the slope of $\rho(L_i t)$ is. Then take the limit as t goes to zero. Recall that for any instant in both ρ and L_i are zero; they become non-zero only when aggregated over some finite length of time.

degree one in all inputs, rather it implies $\rho(\lambda L) = 1 - [1 - \rho(L)]^2$. Eq. (15) satisfies this condition.

Duplicability implies that while the total amount of R&D undertaken in a given industry will be uniquely determined, its distribution across firms within the industry is not. Lambson and Phillips (2005) show that the incentives for R&D depend on how monopoly profits are divided in the case of R&D ties. If they are maintained and split among successful innovators, because of collusion for example, the number of firms will be large and each firm will be very small. If profits are dissipated by Bertrand competition in the case of ties, the number and size of firms are indeterminate, but the industry will switch randomly between competitive and monopoly production levels. To avoid these complications, we assume there is only one R&D firm per industry. Examination of aggregate behavior with Bertrand competition or collusion in the case of ties is an area for future research.

Our model exhibits ‘scale effects’ common to simple general equilibrium models of growth driven by R&D.¹⁶ In our model these scale effects are apparent in that the long-run growth rate depends upon the amount of labor in R&D which in turn depends on the size of the labor force. Scale effects might also manifest themselves in short-run behavior via endogenous labor supply movements driven by the labor–leisure decisions of households. However, since we keep the size of the labor force constant and do not consider a labor–leisure tradeoff, the potential scale effects are never realized. Alternatively, we could model the labor force as growing at some constant rate and eliminate the scale effects. Jones (1995), Kortum (1997), Segerstrom (1998) and Young (1998) propose various ways to eliminate scale effects. In our view the most intuitively appealing method is Segerstrom’s which models the difficulty of doing R&D as rising with the level of technology.¹⁷ In our case this would mean rewriting (15) as $\rho_i = 1 - \exp\{-\kappa(L_i/A)\}$. We transform the model below by dividing all growing variables by A . Hence, adopting this formulation would yield the same stationary simulations as our simple model with fixed labor. The major substantive difference is that real wages would remain constant in the steady state, whereas they grow at the same rate as technology in our model. Also, as noted previously, our model cannot address the comovements of employment and labor force participation with other macroeconomic variables.

In a symmetric equilibrium, arising when all firms have the same probability function for ρ_i , the reward for success in (14) will be the same for all firms. All firms then face an identical problem. The solution to the problem is for the firm to hire the amount of labor it can afford, given the constraint from equity sales. Consequently, $q_i^R = wL_i$ for each industry and aggregate employment by all research firms is

¹⁶See Jones (1995) for a discussion of this issue.

¹⁷Any ‘semi-endogenous’ growth model has similar implications. These models all imply the growth rate of technology is proportional to the exogenous growth rate of labor. They also imply that, ceteris paribus, economies with larger labor forces will have higher levels of technology. Allowing capital in the R&D process is one way to eliminate the latter, however that degree of complexity is beyond the scope of this paper.

$L = IL_i$. In addition, since the expected revenue streams for all intermediate goods producers are the same, the prices of equities for all research and production firms are equal.

2.5. Market-clearing conditions

In addition to the Euler equations from the household and firms' problems, market clearing conditions must be satisfied. Clearing of the labor and capital markets requires

$$\sum_{i=1}^I [L_i + N_i] = L + N = 1, \quad (16)$$

$$\sum_{i=1}^I K_i = K, \quad (17)$$

and clearing of equity markets requires

$$P = P' = 1, \quad (18)$$

$$R = R' = 1. \quad (19)$$

With these conditions, Walras' law ensures goods market clearing.

3. The transformed model

The model economy experiences growth in consumption and output per household due to the increases in A over time. It will be convenient to work with a transformed model where the endogenous variables are all stationary. Since transformations to induce stationarity are commonly used, we delegate details to Appendix B and hereafter consider a transformed version of the model. Variables growing at the same rate as A are transformed with division by A , and the transformed variables will be denoted with a caret so that, for example, $\hat{K} \equiv K/A$.

The transformed law of motion for A is

$$g'_A \equiv A'/A = 1 + \rho(\theta - 1) + \varepsilon'. \quad (20)$$

where $\varepsilon = [(J/I) - \rho](\theta - 1)$ and J is distributed Binomial (I, ρ) so that $E\{\varepsilon\} = 0$, and $\text{Var}\{\varepsilon\} = (1/I)\rho(1 - \rho)(\theta - 1)^2$.

The law of motion for z remains

$$z' = \psi z + \eta' \quad (21)$$

with η' distributed $N(0, \sigma^2)$.

We use a standard CES momentary utility function of the form, $u(C) = (1/1 - \gamma)(C^{1-\gamma} - 1)$. Substituting for marginal utility and transforming household

optimality conditions (1)–(3) gives

$$\begin{aligned} E\{\hat{C}^{-\gamma}\}\hat{q}^P &= \beta(1-\rho)E\left\{\left[1 + \left(\rho \frac{I-1}{I} + \frac{1}{I}\right)(\theta-1) + \frac{I-1}{I} \varepsilon'\right]^{1-\gamma} \right. \\ &\quad \left. \times \hat{C}'^{-\gamma}(\hat{\pi}' + \hat{q}^{P'})\right\}, \end{aligned} \quad (22)$$

$$E\{\hat{C}^{-\gamma}\}\hat{q}^R = \beta\rho E\left\{\left[1 + \rho \frac{I-1}{I}(\theta-1) + \frac{I-1}{I} \varepsilon'\right]^{1-\gamma} \hat{C}'^{-\gamma}(\hat{\pi}' + \hat{q}^{P'})\right\}, \quad (23)$$

$$E\{\hat{C}^{-\gamma}\} = \beta[1 + \rho(\theta-1) + \varepsilon']^{-\gamma} E\{\hat{C}'^{-\gamma}\}(1-\delta+r'). \quad (24)$$

Substituting (16), (18) and (19) into the household's budget constraint and taking expectations gives

$$\hat{C} = \hat{w} + (1-\delta+r)\hat{K} + \hat{\pi} - \hat{K}'g'_A - \hat{q}^R. \quad (25)$$

The additional transformed equations from firms' decisions are as follows:

$$\hat{Y} = \hat{K}^\alpha [e^z(1-L)]^{1-\alpha}, \quad (26)$$

$$\hat{w}(1-L) = \frac{1-\alpha}{\theta^{1-\alpha}} \hat{Y}, \quad (27)$$

$$r\hat{K} = \frac{\alpha}{\theta^{1-\alpha}} \hat{Y}, \quad (28)$$

$$\hat{\pi} = \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha}} \hat{Y}, \quad (29)$$

$$\rho = 1 - \exp\{-\kappa L\}, \quad (30)$$

$$\hat{q}^R = \hat{w}L. \quad (31)$$

Eq. (20)–(31) define the dynamic model that we solve, parameterize, and simulate. Because the highly non-linear nature of the system makes closed form solutions intractable, we consider a linear approximation of the system about its steady state.

4. Calibration and simulation

We solve the system using the method of undetermined coefficients presented in [Christiano \(2002\)](#). Because we approximate about the model's steady state, we first need to solve for the steady state. The parameters and their values are listed in [Table 2](#). There are six parameters which define the steady state: α , β , γ , δ , κ , and θ . For consistency with existing RBC-model parameterizations, we set capital's share in output, α , to 0.3, the quarterly discount factor, β , to 0.995, and the quarterly

Table 2
Values of parameters used in simulations

Parameter	Description	Value
α	Capital share in GDP	0.30
β	Time discount factor for utility	0.995
δ	Depreciation rate	0.02
θ	Size of rungs on the applied technology ladder	1.05162
γ	Elasticity of substitution	0.1397
κ	Ease of R&D	3.390
ψ	Autocorrelation for z shocks	0.95
I	Number of intermediate goods sectors	1, 4, 1,000,000
σ	Standard deviation of z innovations	0, 0.0095, 0.019

depreciation rate, δ , to 0.02. We also set the autocorrelation coefficient on the z shocks, ψ , to 0.95.

θ , the jump up the technology ladder, is set to 1.05162, a value which sets the variance of output to the observed U.S. value. κ and γ are chosen to exactly fix two steady state values, the average quarterly growth rate of real output, and the user cost of capital, $r - \delta$. We set average quarterly real growth equal to 0.00834, and $r - \delta$ to 0.0062, or 2.5% per annum, by setting κ and γ equal the appropriate values.¹⁸ Given the parameter values we have assigned, the steady state probability of success in applied R&D is $\rho = 0.16$.

The standard deviation of technology shocks is set at 0.019, again to generate the observed volatility in output. This value is similar to those used in most RBC analyses. For example, using data from the U.S. economy in the post Bretton Woods era, Schlagenhauf and Wrase (1995) use a value of 0.014. Our objective is to see how the model, with substantially less reliance on auto-correlated productivity shocks than standard RBC models, performs in accounting for business cycle fluctuations observed in key macroeconomic variables. Given the parameter values we use, the standard deviation of the innovations to R&D in (20) is 0.020 when $I = 1$. We are not, however, simply assuming the variability of the R&D innovations. The variability of innovations to the evolution of A in (20) depends on the probability of success in R&D. In turn, the success probability depends endogenously on labor choices made by R&D firms and households in the model. We cannot simply choose the standard deviation of innovations in applied R&D without restriction. One restriction is that the probability of success in the model must be an empirically plausible value.

Note that the model generates a series of growth rates. We can use these to construct a series for the level of technology, the A 's, and then convert all the

¹⁸To see how γ is chosen, consider the steady-state version of Eq. (24): $1 = \beta(\bar{g}'_A)^{\bar{\gamma}}(1 - \delta + \bar{r})$. Note that it is necessary to pick a value of β sufficiently large if we are to generate positive values for γ .

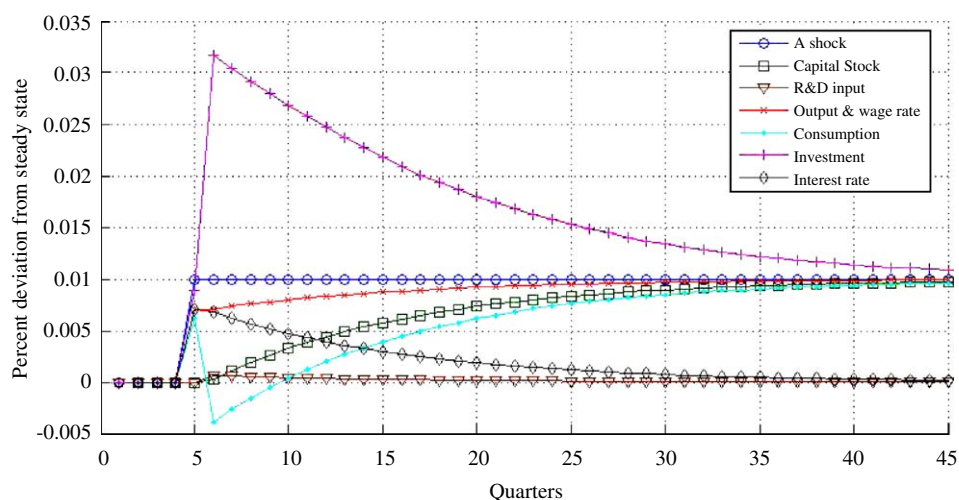
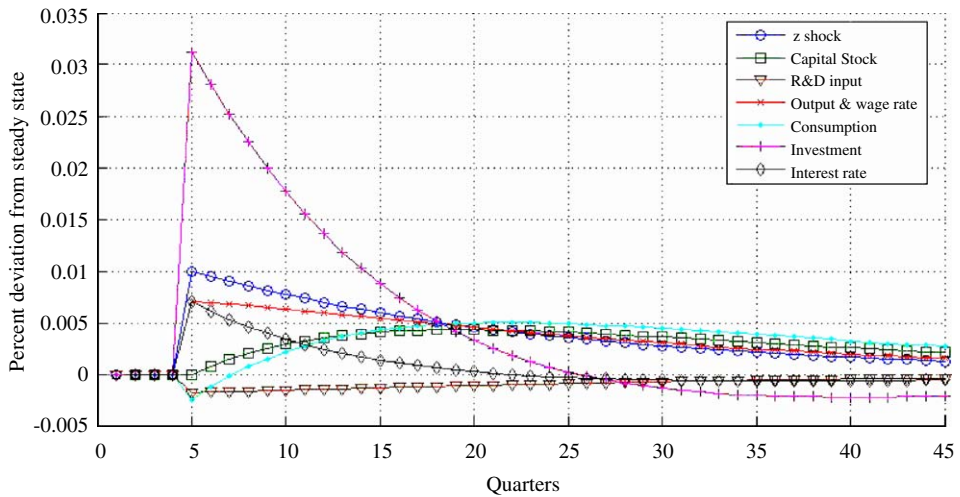


Fig. 1. Impulse responses to 1% shock to A .

stationary, caret-bearing, variables to their non-stationary counterparts. Hence, simulation of the model generates data with both cyclical and growth components.

The model was simulated 10,000 times using a sample of 224 observations, corresponding to our U.S. data set. We filter both actual data drawn from the U.S. economy and model-generated data using two filters: the HP filter; and to consider movements of variables at other than simply business cycle frequencies, [Baxter and King's \(1999\)](#) band-pass filter. We choose three frequency bands. The first emphasizes fluctuations with a periodicity of two to six quarters, a frequency substantially higher than that usually associated with business cycles. The second band is 6–32 quarters, which corresponds closely to the frequencies emphasized by the HP-filter. Finally, we use a band of 20–80 quarters, which emphasizes cycles that last much longer than business cycles, more closely corresponding to Schumpeter's medium-range cycles. We then compute statistical properties of the data, and compare properties of model-generated data with like properties of actual data. We focus on standard RBC measures of variability, cyclicality, and persistence, but consider more than the business cycle frequencies that are the sole focus of RBC analyses.

The responses of endogenous variables to both types of shocks are illustrated by the impulse response functions in [Fig. 1](#) and [2](#). These are also derived without reliance on any particular filtering method. We first simulate the steady state and then impose 1% shocks on both A and z . The impulse responses show the percent deviations of the resulting non-stationary series from their steady state values. Responses to Schumpeterian shocks show permanent changes, but also greater short-run movement in consumption and R&D labor than responses to exogenous shocks.

Fig. 2. Impulse responses to 1% shock to z .

5. Quantitative results

Table 3 presents business cycle moments summarizing behaviors of key macroeconomic aggregates for the U.S. economy over a sample period 1947:I to 2002:IV.¹⁹ We define output as real GDP, consumption as real consumption of services and non-durables, and investment as real investment in non-residential structures and equipment. As is well known, and as revealed in the table, investment is much more variable than output, with variability measured by standard deviation. Consumption has only three-quarters the variability of output. In addition, output is highly serially correlated, consumption is highly correlated with output contemporaneously and at one to two period leads and lags, and investment is not strongly correlated with output contemporaneously or at leads and lags.

The purpose of this paper is not necessarily to match all of these business cycle facts. While it would certainly be desirable to do so, our primary question is how well a model with endogenized shocks from a Schumpeterian framework performs relative to standard exogenous-shock RBC models. It is well-known that simple versions of such models cannot replicate all business cycle facts and require additional modeling details. With the exception of the endogenous R&D process, our model is quite basic and lacks many features needed to conform to measured movements. To answer our fundamental question, therefore, we proceed to solve and simulate several different versions of our model and compare them.

¹⁹The data are real GDP, real consumption of services and non-durables, and real gross fixed investment in non-residential structures and equipment.

Table 3
Moments from U.S. data (1947:I–2002:IV)

	<i>Y</i>	<i>C</i>	<i>I</i>		<i>Y</i>	<i>C</i>	<i>I</i>
(HP filter ($\lambda = 1600$))				Band pass (6,32)			
Standard deviation	0.0175	0.0135	0.1025	Standard deviation	0.0149	0.0121	0.0759
Relative to <i>Y</i>	1	0.7724	5.8523	Relative to <i>Y</i>	1	0.8140	5.1020
Steepness	−0.6060	−0.2555	0.1292	Steepness	−0.4496	−0.3064	0.3299
Deepness	−0.0847	−0.2262	0.0004	Deepness	−0.2630	0.2352	1.3217
Correlation with <i>Y</i> + 4	0.0747	0.2480	0.0107	Correlation with <i>Y</i> + 4	0.0399	0.1787	0.1318
Correlation with <i>Y</i> + 3	0.3249	0.3997	0.2105	Correlation with <i>Y</i> + 3	0.3373	0.4087	0.3317
Correlation with <i>Y</i> + 2	0.6030	0.5494	0.4047	Correlation with <i>Y</i> + 2	0.6568	0.6140	0.5252
Correlation with <i>Y</i> + 1	0.8449	0.6319	0.5310	Correlation with <i>Y</i> + 1	0.9054	0.7146	0.6123
Correlation with <i>Y</i>	1	0.5880	0.5382	Correlation with <i>Y</i>	1	0.6545	0.5217
Correlation with <i>Y</i> − 1	0.8450	0.3698	0.2738	Correlation with <i>Y</i> − 1	0.9054	0.4394	0.2704
Correlation with <i>Y</i> − 2	0.6020	0.0964	−0.0400	Correlation with <i>Y</i> − 2	0.6568	0.1293	−0.0610
Correlation with <i>Y</i> − 3	0.3238	−0.1774	−0.3093	Correlation with <i>Y</i> − 3	0.3373	−0.1830	−0.3595
Correlation with <i>Y</i> − 4	0.0758	−0.3532	−0.4781	Correlation with <i>Y</i> − 4	0.0399	−0.4081	−0.5298
Band pass (2,6)				Band pass (20,80)			
Standard deviation	0.0041	0.0045	0.0336	Standard deviation	0.0133	0.0084	0.0365
Relative to <i>Y</i>	1	1.0819	8.1586	Relative to <i>Y</i>	1	0.6335	2.7418
Steepness	−0.2638	0.1477	−0.0905	Steepness	−0.0763	0.0329	0.2348
Deepness	0.2261	−0.1537	−0.0754	Deepness	−0.6442	−0.7053	−0.6322
Correlation with <i>Y</i> + 4	0.1619	0.2376	0.0341	Correlation with <i>Y</i> + 4	0.6288	0.6416	0.5880
Correlation with <i>Y</i> + 3	−0.1730	−0.1468	0.0344	Correlation with <i>Y</i> + 3	0.7754	0.6968	0.6141
Correlation with <i>Y</i> + 2	−0.4161	−0.1535	−0.3033	Correlation with <i>Y</i> + 2	0.8922	0.7158	0.6011
Correlation with <i>Y</i> + 1	−0.0955	−0.0079	−0.1409	Correlation with <i>Y</i> + 1	0.9695	0.6974	0.5461
Correlation with <i>Y</i>	1	0.2879	0.5004	Correlation with <i>Y</i>	1	0.6435	0.4516
Correlation with <i>Y</i> − 1	−0.0955	0.0093	−0.0243	Correlation with <i>Y</i> − 1	0.9695	0.5451	0.3216
Correlation with <i>Y</i> − 2	−0.4161	−0.0613	−0.0808	Correlation with <i>Y</i> − 2	0.8922	0.4222	0.1774
Correlation with <i>Y</i> − 3	−0.1730	0.0028	−0.1034	Correlation with <i>Y</i> − 3	0.7754	0.2867	0.0324
Correlation with <i>Y</i> − 4	0.1619	−0.2520	−0.2141	Correlation with <i>Y</i> − 4	0.6288	0.1503	−0.1007

Source: U.S. Bureau of Economic Analysis.

The first model we simulate is a simple RBC model with no R&D process and a fixed supply of labor. Here there is only one source of shocks. In all other respects we treat the model exactly as the one outlined above. We assume a fixed rate of growth for *A*, rather than a stochastic one, for example, and add this trend into the simulated model when creating artificial data series. These series are then detrended using the same filters as Table 3. The results are reported in Table 4.²⁰ The table shows the usual sorts of problems with such simple models. The volatilities of consumption and investment are too low. The auto-correlations of output over time are not too different from the U.S. data, but the correlations of consumption and investment with leads and lags of output are very poor matches regardless of the filtering method used. We also report measures of business cycle asymmetry, since

²⁰Note that asterisks denote rejection of the null hypothesis that the simulated moments are the same as the moments for the real world data. Unlike reports of regression coefficients, rejection indicates bad model fit.

Table 4
Moments from a pure exogenous shock version of our model (Model 1): $\sigma = 0.019$

	Y	C	I	Y	C	I
HP filter ($\lambda = 1600$)						
Standard deviation	0.0175	0.0092***	0.0644***	0.0145	0.0078***	0.0532***
Relative to Y	1.0000	0.5262***	3.6892***	1.0000	0.5375***	3.6629***
Steepness	-0.0008***	0.0160	-0.2215	-0.0024	-0.0104	-0.1897*
Deepness	-0.0013	-0.1250	-0.0164	-0.0020	-0.0873	-0.0168***
Correlation with Y + 4	0.1162	-0.4177**	0.2264**	0.0341	-0.4315***	0.1646
Correlation with Y + 3	0.2783	-0.4343***	0.3685	0.3061	-0.5284***	0.4264
Correlation with Y + 2	0.4776*	-0.4329***	0.5365*	0.6270	-0.5790***	0.7156***
Correlation with Y + 1	0.7177***	-0.4091***	0.7322***	0.8930	-0.5192***	0.9237***
Correlation with Y	1.0000	-0.3568***	0.9552	1.0000	-0.3155***	0.9515***
Correlation with Y - 1	0.7177***	0.0780	0.5847***	0.8930	-0.0048***	0.7606***
Correlation with Y - 2	0.4776	0.3724***	0.2923***	0.6270	0.3516***	0.4258***
Correlation with Y - 3	0.2783	0.5530***	0.0688***	0.3061	0.6280***	0.0660***
Correlation with Y - 4	0.1161	0.6443***	-0.0965***	0.0341	0.7721***	-0.2111***
Band pass (2.6)						
Standard deviation	0.0067***	0.0026***	0.0263***	0.0137	0.0093	0.0446
Relative to Y	1.0000	0.3949***	3.9443***	1.0000	0.6820	3.2582***
Steepness	-0.0007	0.0503	-0.0868	-0.0081	-0.0234	-0.0975
Deepness	0.0005	-0.0200	-0.0007	-0.0004	-0.0139	0.0006
Correlation with Y + 4	0.1809	-0.1027***	0.1652*	0.6563	-0.4150***	0.8126***
Correlation with Y + 3	-0.0872	0.1686***	-0.1051	0.7919	-0.3097***	0.8954***
Correlation with Y + 2	-0.3906	0.3525***	-0.3848	0.8995	-0.1789***	0.9404***
Correlation with Y + 1	-0.2104*	0.0544	-0.1785	0.9711	-0.0271***	0.9417***
Correlation with Y	1.0000	-0.9354***	0.9924***	1.0000	0.1399***	0.8955***
Correlation with Y - 1	-0.2104*	0.3975***	-0.2516***	0.9711	0.3111***	0.7922***
Correlation with Y - 2	-0.3906	0.4225***	-0.3998***	0.8995	0.4755	0.6511***
Correlation with Y - 3	-0.0872	0.0233	-0.0741	0.7919	0.6234***	0.4828***
Correlation with Y - 4	0.1809	-0.2089	0.1879***	0.6563	0.7469***	0.2985***
Band pass (20.80)						
Standard deviation						
Relative to Y						
Steepness						
Deepness						
Correlation with Y + 4						
Correlation with Y + 3						
Correlation with Y + 2						
Correlation with Y + 1						
Correlation with Y						
Correlation with Y - 1						
Correlation with Y - 2						
Correlation with Y - 3						
Correlation with Y - 4						

*Significantly different from the corresponding U.S. data moment at 90% confidence.

**Significantly different from the corresponding U.S. data moment at 95% confidence.

***Significantly different from the corresponding U.S. data moment at 99% confidence.

the shocks from the A process have the potential of being highly skewed, especially in our parameterization of the model. We use Sichel's (1993) measures of deepness and steepness, which are the skewness of the series and the skewness of its first-difference, respectively. In this case, of course, the shocks are not skewed, since the innovations to z are normally distributed, and with a few exceptions, the deepness and steepness measures are not significantly different from the U.S. measures.

Our next attempt is to see how well a model driven only by the Schumpeterian source of shocks does in fitting the U.S. moments. The results of this model, where we set I to the extreme value of one, and σ equal to zero, are presented in Table 5. As that table shows, the fit is worse than the simple exogenous shock model, though not considerably so. The volatility of consumption is extremely high when high frequency filtering methods are used. In addition, the model matches none of the business cycle asymmetries. In terms of other moments, it does no better or worse than the exogenous shock case.

We next consider a version of our model driven entirely by exogenous shocks. This is not the same as the model in Table 4, however, since our model allows for these exogenous shocks to influence the allocation of labor between production and R&D. The results of setting I equal to 1 million and σ to the same value as Table 4, are reported in Table 6. The only substantive difference this model yields is a drop in the volatility of investment at low frequencies.

Finally, we consider a version of the model where shocks come from both sources. We set I to four (giving half the volatility from A as before) and σ to 0.0161 (the amount of additional volatility needed to match output volatility with the U.S. data). The exogenous shocks are the most important source of variation, but the contribution of endogenous shocks is still non-trivial. These results are in Table 7. Consumption volatility is too high and investment volatility is too low. This model does as poorly as the others in matching correlations with leads and lags of output. The overall fit is not overwhelmingly better or worse.

In order to formalize our comparison of moments across models, we compute measures of the percentage deviations of model moments from the U.S. data moments. We report both the root mean squared deviation (RMSD), and the mean absolute deviations (MAD) for the volatility of output, the relative volatility of consumption, the relative volatility of investment, and the contemporaneous correlations of consumption and investment with output. These are reported in Table 8. Our model improves on the simple RBC model, though the best fit is not always the same version. Using the HP-filter, the model with Schumpeterian shocks only has the best fit. For the band-pass filter, the version with both types of shocks is much better at matching high frequency moments, while the model with only z shocks is best at lower frequencies.

The model does not do well in matching business cycle asymmetries. For the U.S., drops in output tend to be bigger and more sudden than increases. This shows up as a negative value for the steepness measure of -0.6060 and a deepness value close to zero. Schumpeterian shocks, on the other hand, tend to do the opposite. This shows up as steepness measures that are closer to zero and deepness measures that are strongly positive.

Table 5
Moments from our model driven only by $\mathcal{A}$ shocks (Model 2): $I = 1, \sigma = 0$

	Y	C	I	Y	C	I
HP filter ($\lambda = 1600$)						
Standard deviation	0.0172	0.0208***	0.0648***	0.0143	0.0112	0.0551***
Relative to Y	1.0000	1.2152***	3.7769***	1.0000	0.7891	3.8664***
Steepness	-0.1520***	0.4408***	0.1126***	-0.1508***	-0.5813***	0.0591***
Deepness	1.8584***	-0.8024***	1.8823***	0.5385***	-0.1910***	0.5164***
Correlation with $Y + 4$	0.1420	-0.0944**	0.1631	0.0721	-0.0079	0.0544
Correlation with $Y + 3$	0.3006	-0.0955***	0.2847	0.3350	0.0263***	0.2358
Correlation with $Y + 2$	0.4919	-0.0931***	0.4294	0.6429	-0.0484***	0.4935
Correlation with $Y + 1$	0.7171***	-0.0852***	0.5968	0.8976	-0.2504***	0.7622***
Correlation with Y	1.0000	-0.0476***	0.7902***	1.0000	-0.5013***	0.9380***
Correlation with $Y - 1$	0.7171***	-0.6282***	0.9318***	0.8976	-0.6530***	0.9243***
Correlation with $Y - 2$	0.4919	-0.3055***	0.5677***	0.6429	-0.5956***	0.7163***
Correlation with $Y - 3$	0.3006	-0.0650**	0.2779**	0.3350	-0.3215*	0.3838***
Correlation with $Y - 4$	0.1420	0.1059***	0.0538	0.0721	0.0643***	0.0399***
Band pass (2.6)						
Standard deviation	0.0066***	0.0160***	0.0225***	0.0143	0.0098	0.0484
Relative to Y	1.0000	2.4126***	3.4120***	1.0000	0.6848	3.3967***
Steepness	-0.0091***	0.7817***	0.3427***	-0.0641***	-0.1200***	0.0284***
Deepness	0.9774***	-0.3935***	1.0042***	0.0558***	0.1894***	-0.0219***
Correlation with $Y + 4$	0.1755	-0.0553***	0.2192**	0.6825	-0.4210***	0.7406***
Correlation with $Y + 3$	-0.0760	-0.1659	0.1553	0.8079	-0.3969***	0.8388***
Correlation with $Y + 2$	-0.3669	-0.1113	-0.1587	0.9072	-0.3498***	0.9064***
Correlation with $Y + 1$	-0.2249**	0.1750**	-0.4174***	0.9732	-0.2788***	0.9370***
Correlation with Y	1.0000	0.6594***	-0.0372***	1.0000	-0.1844***	0.9257***
Correlation with $Y - 1$	-0.2249**	-0.8731***	0.9679***	0.9732	-0.0685***	0.8606***
Correlation with $Y - 2$	-0.3669	0.0007	-0.3095***	0.9072	0.0707***	0.7534***
Correlation with $Y - 3$	-0.0760	0.2637***	-0.4111***	0.8079	0.2227	0.6128***
Correlation with $Y - 4$	0.1755	0.1486***	-0.0483**	0.6825	0.3773**	0.4486***
Band pass (20.80)						
Standard deviation	0.0066***	0.0160***	0.0225***	0.0143	0.0098	0.0484
Relative to Y	1.0000	2.4126***	3.4120***	1.0000	0.6848	3.3967***
Steepness	-0.0091***	0.7817***	0.3427***	-0.0641***	-0.1200***	0.0284***
Deepness	0.9774***	-0.3935***	1.0042***	0.0558***	0.1894***	-0.0219***
Correlation with $Y + 4$	0.1755	-0.0553***	0.2192**	0.6825	-0.4210***	0.7406***
Correlation with $Y + 3$	-0.0760	-0.1659	0.1553	0.8079	-0.3969***	0.8388***
Correlation with $Y + 2$	-0.3669	-0.1113	-0.1587	0.9072	-0.3498***	0.9064***
Correlation with $Y + 1$	-0.2249**	0.1750**	-0.4174***	0.9732	-0.2788***	0.9370***
Correlation with Y	1.0000	0.6594***	-0.0372***	1.0000	-0.1844***	0.9257***
Correlation with $Y - 1$	-0.2249**	-0.8731***	0.9679***	0.9732	-0.0685***	0.8606***
Correlation with $Y - 2$	-0.3669	0.0007	-0.3095***	0.9072	0.0707***	0.7534***
Correlation with $Y - 3$	-0.0760	0.2637***	-0.4111***	0.8079	0.2227	0.6128***
Correlation with $Y - 4$	0.1755	0.1486***	-0.0483**	0.6825	0.3773**	0.4486***

*Significantly different from the corresponding U.S. data moment at 90% confidence.

**Significantly different from the corresponding U.S. data moment at 95% confidence.

***Significantly different from the corresponding U.S. data moment at 99% confidence.

Table 6
Moments from our model driven only by z shocks (Model 3): $I = 1,000,000$, $\sigma = 0.019$

	Y	C	I		Y	C	I
HP filter ($\lambda = 1600$)							
Standard deviation	0.0175	0.0097***	0.0622***	Band pass (6,32)	0.0146	0.0082***	0.0513***
Relative to Y	1.0000	0.5531***	3.5537***	Standard deviation	1.0000	0.5645***	3.5263***
Steepness	-0.0003***	0.0205***	-0.2125***	Relative to Y	0.0013***	-0.0006***	-0.1799***
Deepness	-0.0014***	-0.1165***	-0.0157***	Steepness	0.0014***	-0.0877***	-0.0150***
Correlation with $Y + 4$	0.1194	-0.4146***	0.2313**	Deepness	0.0385	-0.4298***	0.1711
Correlation with $Y + 3$	0.2813	-0.4294***	0.3723*	Correlation with $Y + 4$	0.3094	-0.5246***	0.4317
Correlation with $Y + 2$	0.4804	-0.4264***	0.5391*	Correlation with $Y + 3$	0.6288	-0.5730***	0.7188***
Correlation with $Y + 1$	0.7193***	-0.4010***	0.7326***	Correlation with $Y + 2$	0.8935	-0.5109***	0.9241***
Correlation with Y	1.0000	-0.3476***	0.9528***	Correlation with $Y + 1$	1.0000	-0.3052***	0.9490***
Correlation with $Y - 1$	0.7193***	0.0892***	0.5801***	Correlation with Y	0.8935	0.0164***	0.7556***
Correlation with $Y - 2$	0.4804	0.3833***	0.2870***	Correlation with $Y - 1$	0.6288	0.3632***	0.4194***
Correlation with $Y - 3$	0.2812	0.5625***	0.0630***	Correlation with $Y - 2$	0.3094	0.6383***	0.0596***
Correlation with $Y - 4$	0.1192	0.6514***	-0.1017***	Correlation with $Y - 3$	0.0385	0.7799***	-0.2165***
Band pass (2,6)							
Standard deviation	0.0067***	0.0028***	0.0255***	Band pass (20,80)	0.0138	0.0098	0.0429
Relative to Y	1.0000	0.4193***	3.8200***	Standard deviation	1.0000	0.7084**	3.1152***
Steepness	0.0003***	0.0451***	-0.0810***	Relative to Y	0.0036***	-0.0150***	-0.0806***
Deepness	-0.0027***	-0.0146***	-0.0042***	Steepness	0.0048***	-0.0162***	-0.0037***
Correlation with $Y + 4$	0.1789	-0.0993***	0.1625*	Deepness	0.6572	-0.4053***	0.8161***
Correlation with $Y + 3$	-0.0885	0.1706***	-0.1071	Correlation with $Y + 4$	0.7924	-0.2982***	0.8972***
Correlation with $Y + 2$	-0.3889	0.3512***	-0.3833	Correlation with $Y + 3$	0.8998	-0.1659***	0.9402***
Correlation with $Y + 1$	-0.2100*	0.0536	-0.1770	Correlation with $Y + 2$	0.9712	-0.0127***	0.9394***
Correlation with Y	1.0000	-0.9359***	0.9927***	Correlation with $Y + 1$	1.0000	0.1551***	0.8912***
Correlation with $Y - 1$	-0.2100*	0.3997***	-0.2532***	Correlation with Y	0.9712	0.3264***	0.7860***
Correlation with $Y - 2$	-0.3889	0.4222***	-0.3989***	Correlation with $Y - 1$	0.8998	0.4903***	0.6434***
Correlation with $Y - 3$	-0.0885	0.0245	-0.0749	Correlation with $Y - 2$	0.7924	0.6371***	0.4739***
Correlation with $Y - 4$	0.1789	-0.2079	0.1865***	Correlation with $Y - 3$	0.6572	0.7588***	0.2889***
Correlation with $Y - 4$				Correlation with $Y - 4$			

*Significantly different from the corresponding U.S. data moment at 90% confidence.

**Significantly different from the corresponding U.S. data moment at 95% confidence.

***Significantly different from the corresponding U.S. data moment at 99% confidence.

Table 7
Moments from our model driven by a combination of A & z shocks (Model 4): $I = 4$, $\sigma = 0.0161$

	Y	C	I		Y	C	I
HP filter ($\lambda = 1600$)							
Standard deviation	0.0173	0.0121	0.0624***	Band pass (6,32)	0.0144	0.0087***	0.0523***
Relative to Y	1.0000	0.7056	3.6195***	Standard deviation	1.0000	0.6105***	3.6413***
Steepness	-0.0096***	0.0829***	-0.1915***	Relative to Y	-0.0081***	-0.0597***	-0.1675***
Deepness	0.1102***	-0.4535***	0.0658***	Steepness	0.0295***	-0.0706***	0.0187***
Correlation with $Y + 4$	0.1250	-0.2967***	0.2192*	Deepness	0.0458	-0.3196***	0.1453
Correlation with $Y + 3$	0.2863	-0.3076***	0.3567	Correlation with $Y + 4$	0.3151	-0.3834***	0.3877
Correlation with $Y + 2$	0.4836	-0.3060***	0.5191	Correlation with $Y + 3$	0.6319	-0.4438***	0.6680**
Correlation with $Y + 1$	0.7195***	-0.2891***	0.7076***	Correlation with $Y + 2$	0.8944	-0.4572***	0.8879***
Correlation with Y	1.0000	-0.2462***	0.9232***	Correlation with $Y + 1$	1.0000	-0.3762***	0.9473***
Correlation with $Y - 1$	0.7195***	-0.1818***	0.6651***	Correlation with Y	0.8944	-0.1795***	0.7947***
Correlation with $Y - 2$	0.4835	0.1233	0.3520***	Correlation with $Y - 1$	0.6319	0.0958	0.4871***
Correlation with $Y - 3$	0.2861	0.3258***	0.1107***	Correlation with $Y - 2$	0.3151	0.3788***	0.1325***
Correlation with $Y - 4$	0.1247	0.4467***	-0.0692***	Correlation with $Y - 3$	0.0458	0.5937***	-0.1601***
Band pass (2,6)							
Standard deviation	0.0066***	0.0069***	0.0240***	Band pass (20,80)	0.0138	0.0097	0.0444
Relative to Y	1.0000	1.0475	3.6429***	Standard deviation	1.0000	0.7080	3.2217***
Steepness	-0.0023***	0.4211***	-0.0614***	Relative to Y	-0.0062***	-0.0189***	-0.0851***
Deepness	0.0589***	-0.2420***	0.0308***	Steepness	0.0037***	-0.0082***	0.0033***
Correlation with $Y + 4$	0.1802	-0.0629***	0.1837***	Deepness	0.6641	-0.4161***	0.7982***
Correlation with $Y + 3$	-0.0853	-0.0269	-0.0551	Correlation with $Y + 4$	0.7966	-0.3287***	0.8825***
Correlation with $Y + 2$	-0.3851	0.0635**	-0.3530	Correlation with $Y + 3$	0.9017	-0.2163***	0.9306***
Correlation with $Y + 1$	-0.2128*	0.1073	-0.2359	Correlation with $Y + 2$	0.9717	-0.0820***	0.9368***
Correlation with Y	1.0000	0.0131***	0.8170***	Correlation with $Y + 1$	1.0000	0.0693***	0.8972***
Correlation with $Y - 1$	-0.2128*	-0.3014***	-0.0060	Correlation with Y	0.9717	0.2288***	0.8015***
Correlation with $Y - 2$	-0.3851	0.1411**	-0.3975***	Correlation with $Y - 1$	0.9017	0.3882	0.6674***
Correlation with $Y - 3$	-0.0853	0.1387	-0.1480	Correlation with $Y - 2$	0.7966	0.5376***	0.5051***
Correlation with $Y - 4$	0.1802	0.0055	0.1458	Correlation with $Y - 3$	0.6641	0.6684***	0.3256***

*Significantly different from the corresponding U.S. data moment at 90% confidence.

**Significantly different from the corresponding U.S. data moment at 95% confidence.

***Significantly different from the corresponding U.S. data moment at 99% confidence.

Table 8

Comparison of fit various models based on five key moments ($\sigma_Y, \sigma_C/\sigma_Y, \sigma_C/\sigma_Y, \rho_{YC}, \rho_{YT}$)

	Model 1: RBC	Model 2: A only	Model 3: z only	Model 4: both
HP filter ($\lambda = 1600$)				
RMSD	0.9409	0.7354	0.9336	0.8467
MAD	0.4595	0.4051	0.4536	0.3801
Band pass (2,6)				
RMSD	2.1619	1.0180	2.1619	0.7127
MAD	1.2059	0.7429	1.2044	0.4291
Band pass (6,32)				
RMSD	0.8802	0.9945	0.8725	0.9166
MAD	0.4257	0.4157	0.4209	0.4295
Band pass (20,80)				
RMSD	0.5361	0.7760	0.5233	0.5889
MAD	0.2152	0.3363	0.2099	0.2441

6. Conclusions

In this paper, we investigate whether the process of creative destruction used in Schumpeterian models of growth is a reasonable source for the technology shocks used to drive business cycles, particularly in the context of RBC models. We conclude that the Schumpeterian process alone is not sufficient. In our simple model with no labor–leisure choice and no diffusion mechanism for these technology shocks, we find that in order to generate realistic volatilities for output we also generate counterfactual volatilities of consumption and investment, and we fail to match observed business cycle asymmetries.

We also show that various versions of our model can generate volatilities, covariances, and persistence properties that match the U.S. experience as least as well as an equivalent RBC model. Even when the aggregate Schumpeterian shocks are smoothed away there is a better fit. This is evidence that the internal propagation mechanism is an improvement over a standard RBC model.

We conclude that further exploration of the potential contribution of Schumpeterian shocks to business cycle and other aggregate fluctuations is worth considering. While this model cannot fit the U.S. moments, it is likely that this failure can be attributed in large part to the lack of sophistication in modeling the non-Schumpeterian aspects of the model. In particular, we do not have any sort of labor-leisure decision in our model. In addition, the model does not allow for the use of capital in R&D. In a growing economy, such features can introduce scale effects which imply explosive growth with no steady state. However the scale effects growth literature offers a wide variety of modeling choices for eliminating this problem. Another area for additional realism would be to make the size technological improvements stochastic. This could be done by simply making θ a random variable independent of the shocks already in the model. Or, firms might be allowed to choose

the expected size of θ with larger jumps being more costly. Again, the growth literature offers guidelines as to how these improvements might be incorporated into the model. Finally, our model does not incorporate sophisticated R&D spillovers or technological diffusion of innovations across sectors. These may well be important and will undoubtedly change the aggregate behavior of output, investment and consumption.

Because of this lack of diffusion we are forced to drive changes in aggregate output by assuming a small number of intermediate goods sectors. It may be more profitable to build a model with a large number of intermediate sectors generating approximately smooth growth, and consider innovations which affect all industries equally as coming from some other source. In earlier related work, we were able to generate more realistic models when we allowed for movements in basic research to be distinct from movements in applied research, for example.

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Appendix A

The reward for success can be written as the expected present value of the profit stream, where this stream is discounted both by the real interest rate and by the probability of losing the monopoly to a newly successful research firm:

$$V'_i = \pi'_i + q^{p'}_i = \sum_{s=t+1}^{\infty} \pi_{i,s} d_{i,s}, \quad d_{i,s} = \prod_{u=1}^s \frac{1 - \rho_{i,u}}{1 + r_{u+1}},$$

where the second subscript, used in the summation and product, indicates the time period.

Future profits are a constant fraction, $(\theta^{1-\alpha} - 1)/\theta^{1-\alpha}$, of production. The level of technology, $\theta' A_i$, does not change as long as the firm is the monopoly producer. Consequently, we can rewrite the success reward as

$$V'_i = \sum_{s=t+1}^{\infty} \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha}} p_{i,s} Y_{i,s} d_{i,s},$$

noting that $p_{i,s}Y_{i,s} = (1/I)Y_s$ gives

$$V'_i = \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha}} \sum_{s=t+1}^{\infty} \frac{1}{I} Y_s d_{i,s} \quad \text{or} \quad V'_i = \frac{\theta^{1-\alpha} - 1}{\theta^{1-\alpha}} M; \quad M \equiv \frac{1}{I} \sum_{s=t+1}^{\infty} Y_s d_{i,s}.$$

Since M changes as aggregate output rises, with large I the expected effects of success or failure by any given research firm on M will be small. Note also that M will be the same for all research firms as long as ρ is the same. There may be more than one solution to the system of equations that jointly determine the M 's and ρ 's for all firms, but one solution is the symmetric one, where M and ρ are constant over time.

Appendix B

In order to induce stationarity, we take all the variables that grow at the same rate as A in the previous section and divide them by A . This gives the following stationary variables:

$$\hat{C} \equiv C/A, \quad \hat{K} \equiv K/A, \quad \hat{w} \equiv w/A, \quad \hat{q}^R \equiv q^R/A,$$

$$\hat{q}^P \equiv q^P/A, \quad \hat{\pi} \equiv \pi/A, \quad g_A \equiv \tilde{A}'/A.$$

The transformed law of motion for A can be written as $g_A \equiv A'/A = \rho(\theta - 1) + 1 + \varepsilon$, where $\varepsilon' = [(J/I) - \rho](\theta - 1)$, and J is distributed Binomial (I, ρ) . Note that $E\{\varepsilon'\} = 0$ and $\text{Var}\{\varepsilon'\} = (\rho(1 - \rho)/I)(\theta - 1)^2$.

Substituting (18) and (19) into the household's budget constraint, and transforming variables, gives: $\hat{C} = \hat{w} + (1 - \delta + r)\hat{K} + \hat{\pi} - \hat{K}'(1 + g_A) - \hat{q}^R$, with expected consumption when the household makes its share decisions given by $E\hat{C} \equiv E\{\hat{C}\} = \hat{w} + (1 - \delta + r)\hat{K} + \hat{\pi} - E\{\hat{K}'(1 + g_A)\} - \hat{q}^R$.

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